

## 17: Updating the data with PreparedStatement

When updating data using PreparedStatement, we can utilize placeholders for both:

- the values we want to update
- the condition for selecting which rows to update

### 👉 Steps for Inserting Data with Prepared Statement:

1. **Create Parameterized Queries:** Use placeholders (?) in the SQL update query for the new values and conditions. This allows dynamic value insertion at runtime, preventing SQL injection and making the query more efficient.

```
String sql = "UPDATE studentinfo SET sage=? WHERE id=?";  
PreparedStatement pstmt = connect.prepareStatement(sql);
```

2. **User input at runtime:** Values for updating can be taken from the user at runtime.

```
Scanner scan = new Scanner(System.in);  
  
System.out.println("Please enter info that needs to be updated");  
  
System.out.println("Kindly enter your id: ");  
  
Integer id = scan.nextInt();  
  
System.out.println("Please enter your age to be updated");  
  
Integer sage = scan.nextInt();
```

3. **Setting Values:** Use the set() methods (e.g., setInt(), setString()) to bind the input values to the placeholders in the SQL query. The first parameter of the set() method is the placeholder index (starting from 1), and the second parameter is the value.

```
pstmt.setInt(1, sage);  
pstmt.setInt(2, id);
```

#### 4. Executing the Update:

The executeUpdate() method is used to execute the update query.

```
int rowsAffected = pstmt.executeUpdate();
```

#### 5. Checking the Result:

After executing the query, you can check if the data was successfully updated by checking the number of affected rows. If the result is greater than 0, it indicates that one or more rows were updated.

```
if(rowsAffected == 0) {  
    System.out.println("Unable to update the data");  
} else {  
    System.out.println("Data Updated Successfully!");  
}
```

### **Code:**

```
public class UpdateStudentInfo {  
  
    public static void main(String[] args) {  
        String url = "jdbc:mysql://localhost:3306/db_name";  
        String username = "root";  
        String password = "password";
```

```

String sql = "UPDATE studentinfo SET sage=? WHERE id=?";

try (
    Connection connect = DriverManager.getConnection(url, username,
password);
    PreparedStatement pstmt = connect.prepareStatement(sql);
    Scanner scan = new Scanner(System.in);
) {
    System.out.println("Please enter info that needs to be updated");
    System.out.print("Kindly enter your ID: ");
    int id = scan.nextInt();
    System.out.print("Please enter your age to be updated: ");
    int sage = scan.nextInt();

    pstmt.setInt(1, sage);
    pstmt.setInt(2, id);

    int rowsAffected = pstmt.executeUpdate();

    if (rowsAffected == 0) {
        System.out.println("Unable to update the data. No rows affected.");
    } else {
        System.out.println("Data Updated Successfully! Rows affected: " +
rowsAffected);
    }

} catch (SQLException e) {
    e.printStackTrace();
}
}
}

```